

Because the degrees of freedom for $SS_{\text{Treatments}}$ and SS_E add to $N - 1$, the total number of degrees of freedom, Cochran's theorem implies that $SS_{\text{Treatments}}/\sigma^2$ and SS_E/σ^2 are independently distributed chi-square random variables. Therefore, if the null hypothesis of no difference in treatment means is true, the ratio

$$F_0 = \frac{SS_{\text{Treatments}}/(a-1)}{SS_E/(N-a)} = \frac{MS_{\text{Treatments}}}{MS_E} \quad (3.7)$$

is distributed as F with $a - 1$ and $N - a$ degrees of freedom. Equation 3.7 is the **test statistic** for the hypothesis of no differences in treatment means.

From the expected mean squares we see that, in general, MS_E is an unbiased estimator of σ^2 . Also, under the null hypothesis, $MS_{\text{Treatments}}$ is an unbiased estimator of σ^2 . However, if the null hypothesis is false, the expected value of $MS_{\text{Treatments}}$ is greater than σ^2 . Therefore, under the alternative hypothesis, the expected value of the numerator of the test statistic (Equation 3.7) is greater than the expected value of the denominator, and we should reject H_0 on values of the test statistic that are too large. This implies an upper-tail, one-tail critical region. Therefore, we should reject H_0 and conclude that there are differences in the treatment means if

$$F_0 > F_{\alpha, a-1, N-a}$$

where F_0 is computed from Equation 3.7. Alternatively, we could use the P -value approach for decision making. The table of F percentages in the Appendix (Table IV) can be used to find bounds on the P -value.

The sums of squares may be computed in several ways. One direct approach is to make use of the definition

$$y_{ij} - \bar{y}_{..} = (\bar{y}_{i.} - \bar{y}_{..}) + (y_{ij} - \bar{y}_{i.})$$

Use a spreadsheet to compute these three terms for each observation. Then, sum up the squares to obtain SS_T , $SS_{\text{Treatments}}$, and SS_E . Another approach is to rewrite and simplify the definitions of $SS_{\text{Treatments}}$ and SS_T in Equation 3.6, which results in

$$SS_T = \sum_{i=1}^a \sum_{j=1}^n y_{ij}^2 - \frac{y_{..}^2}{N} \quad (3.8)$$

$$SS_{\text{Treatments}} = \frac{1}{n} \sum_{i=1}^a y_{i.}^2 - \frac{y_{..}^2}{N} \quad (3.9)$$

and

$$SS_E = SS_T - SS_{\text{Treatments}} \quad (3.10)$$

This approach is nice because some calculators are designed to accumulate the sum of entered numbers in one register and the sum of the squares of those numbers in another, so each number only has to be entered once. In practice, we use computer software to do this.

The test procedure is summarized in Table 3.3. This is called an **analysis of variance** (or **ANOVA**) table.

■ TABLE 3.3

The Analysis of Variance Table for the Single-Factor, Fixed Effects Model

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Between treatments	$SS_{\text{Treatments}} = n \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2$	$a - 1$	$MS_{\text{Treatments}}$	$F_0 = \frac{MS_{\text{Treatments}}}{MS_E}$
Error (within treatments)	$SS_E = SS_T - SS_{\text{Treatments}}$	$N - a$	MS_E	
Total	$SS_T = \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2$	$N - 1$		

EXAMPLE 3.1 The Plasma Etching Experiment

To illustrate the analysis of variance, return to the first example discussed in Section 3.1. Recall that the engineer is interested in determining if the RF power setting affects the etch rate, and she has run a completely randomized experiment with four levels of RF power and five replicates. For convenience, we repeat here the data from Table 3.1:

Usually, these calculations would be performed on a computer, using a software package with the capability to analyze data from designed experiments.

RF Power (W)	Observed Etch Rate ($\text{\AA}/\text{min}$)					Totals $y_{i.}$	Averages $\bar{y}_{i.}$
	1	2	3	4	5		
160	575	542	530	539	570	2756	551.2
180	565	593	590	579	610	2937	587.4
200	600	651	610	637	629	3127	625.4
220	725	700	715	685	710	3535	707.0
						$y_{..} = 12,355$	$\bar{y}_{..} = 617.75$

We will use the analysis of variance to test $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ against the alternative H_1 : some means are different. The sums of squares required are computed using Equations 3.8, 3.9, and 3.10 as follows:

$$\begin{aligned}
 SS_T &= \sum_{i=1}^4 \sum_{j=1}^5 y_{ij}^2 - \frac{y_{..}^2}{N} \\
 &= (575)^2 + (542)^2 + \cdots + (710)^2 - \frac{(12,355)^2}{20} \\
 &= 72,209.75 \\
 SS_{\text{Treatments}} &= \frac{1}{n} \sum_{i=1}^4 y_{i.}^2 - \frac{y_{..}^2}{N} \\
 &= \frac{1}{5} [(2756)^2 + \cdots + (3535)^2] - \frac{(12,355)^2}{20} \\
 &= 66,870.55 \\
 SS_E &= SS_T - SS_{\text{Treatments}} \\
 &= 72,209.75 - 66,870.55 = 5339.20
 \end{aligned}$$

The ANOVA is summarized in Table 3.4. Note that the RF power or between-treatment mean square (22,290.18) is many times larger than the within-treatment or error mean square (333.70). This indicates that it is unlikely that the treatment means are equal. More formally, we can compute the F ratio $F_0 = 22,290.18/333.70 = 66.80$ and compare this to an appropriate upper-tail percentage point of the $F_{3,16}$ distribution. To use a fixed significance level approach, suppose that the experimenter has selected $\alpha = 0.05$. From Appendix Table IV, we find that $F_{0.05,3,16} = 3.24$. Because $F_0 = 66.80 > 3.24$, we reject H_0 and conclude that the treatment means differ; that is, the RF power setting significantly affects the mean etch rate. We could also compute a P -value for this test statistic. Figure 3.3 shows the reference distribution ($F_{3,16}$) for the test statistic F_0 . Clearly, the P -value is very small in this case. From Appendix Table A-4, we find that $F_{0.01,3,16} = 5.29$ and because $F_0 > 5.29$, we can conclude that an upper bound for the P -value is 0.01; that is, $P < 0.01$ (the exact P -value is $P = 2.88 \times 10^{-9}$).

■ **TABLE 3.4**
ANOVA for the Plasma Etching Experiment

Source of Variation	Sum of Square	Degrees of Freedom	Mean Squares	F_0	P -Value
RF Power	66,870.55	3	22,290.18	$F_0 = 66.80$	< 0.01
Error	5339.20	16	333.70		
Total	72,209.75	19			